

UFUG 2104

Assignment 1: Project Report

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1 Introduction

This project studies the dataset of billionaires with the goal of answering a simple question carefully: *what patterns in billionaire wealth are stable enough to report, and what patterns are likely artifacts of data collection?* We start with a data audit (schema freeze and missingness structure) before moving to descriptive statistics (heavy tails and concentration), inferential comparisons (effect sizes with confidence intervals), and regression models linking billionaire wealth to country-level GDP. Throughout, we emphasize **interpretability and robustness**: results are reported on a log scale when appropriate, effect sizes are paired with 95% confidence intervals, and key conclusions are challenged with at least one robustness check.

Billionaire wealth sits at the far end of the economic distribution, where “typical” statistical intuition often fails: the distribution is extremely skewed, a small number of observations can dominate averages, and missingness can be structural rather than random. Because of this, our report is written as a story with guardrails. We begin by asking whether the dataset is *fit for inference* (data quality and coverage), then we describe what the distribution looks like, and only then do we test hypotheses and fit regression models.

1.1 Research Questions

Our research questions (RQs) are summarized in Table 1. Given the complexity and practical significance of the given dataset, we plan to propose research questions that progress from the dataset itself to the relationships among the data. They guide both the analysis and the organization of evidence.

Table 1: Research questions used in this report (English version).

rq_id	question
RQ0	RQ0: Does data quality (missingness/outliers/duplicates) systematically bias certain countries/industries and affect inference?
RQ1	RQ1: Is finalWorth heavy-tailed, and how dominant is the top 1% for means/variance/correlation/regression?
RQ2	RQ2: Do selfMade, gender, and category differ in finalWorth? What are the effect sizes and confidence intervals?
RQ3	RQ3: Is finalWorth associated with macro variables (e.g., GDP), and is the association robust to log transforms and stratification?
RQ4	RQ4: How well does gdp_country predict finalWorth in a simple regression, and are results stable under log, multivariate controls, and robust SE?
RQ5	RQ5: Quantile regression—does the GDP association differ across wealth quantiles?
RQ6	RQ6: Permutation / bootstrap inference to reduce distributional assumptions.

1.2 Brief inspections

In statistical analysis, we employed the the folloeing techniques:

2 Statistical Analysis

2.1 Data Processing and Data Quality Audit

Schema freeze and reproducibility. We treat the raw CSV as immutable input and record a “schema freeze” (column set, read rules, and file fingerprint). This prevents analysis drift and ensures that future reruns of the notebook reproduce the same artifacts.

Missingness is part of the signal. We define the missingness rate of a variable X as

$$\text{MissingRate}(X) = \frac{\#\{i : X_i \text{ is NA}\}}{n}.$$

In this dataset, some missingness reflects *structural coverage* rather than measurement error. For example, Figure 1 shows that `state` and `residenceStateRegion` are populated only for U.S. records. This matters: if we naively compare missingness across countries without this context, we could misinterpret missingness as an economic pattern.

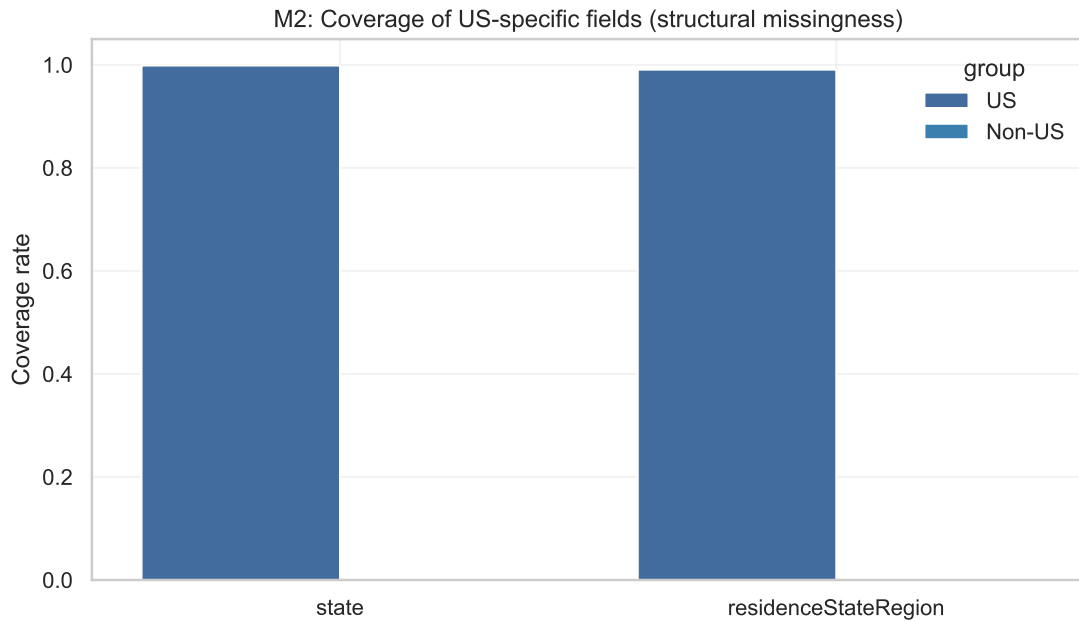


Figure 1: Structural coverage: U.S.-only population for state/region fields.

To keep this report honest, we use missingness summaries as a *filter*: variables with structural missingness are either analyzed within the relevant subset (e.g., U.S. only) or excluded from global models. Figure 2 shows overall missingness, and Figure 3 reveals missingness structure by country (top 12 by frequency).

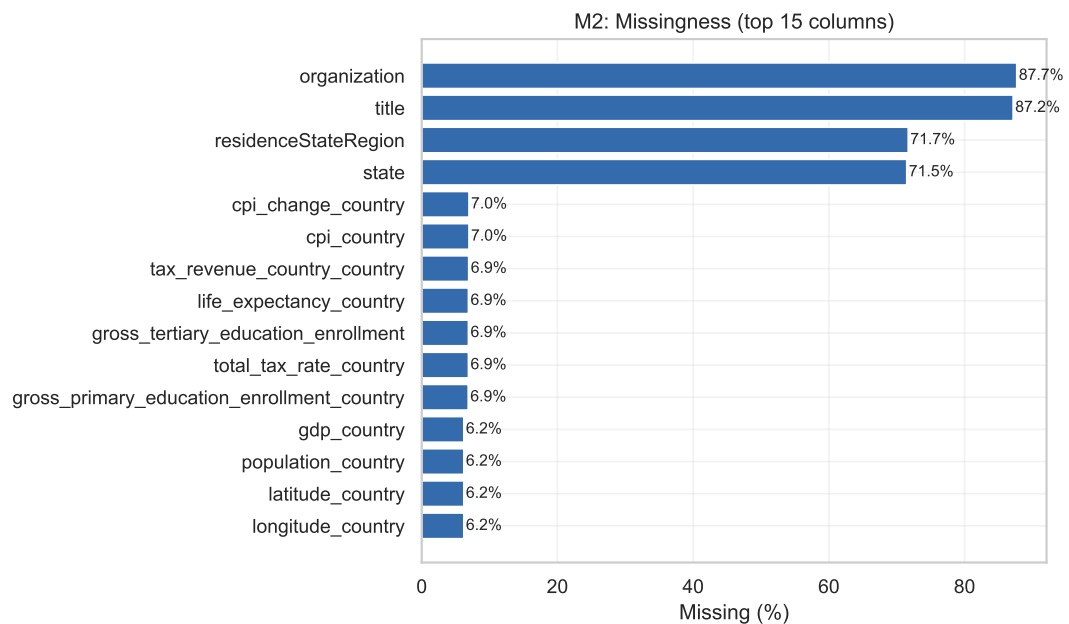


Figure 2: Top-15 variables by missingness rate.

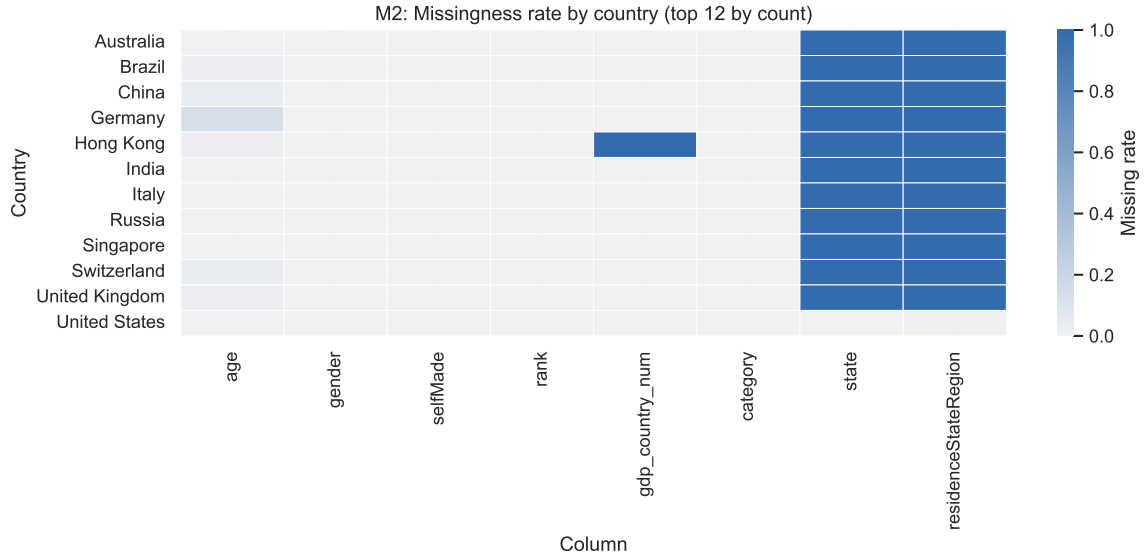


Figure 3: Missingness heatmap by country (top 12 by frequency).

Heavy tails and why we log-transform. On the raw scale, billionaire wealth is so skewed that a small number of extreme observations can dominate summaries and models. We therefore work on the log scale for robustness and interpretability:

$$\log_finalWorth = \log(1 + finalWorth), \quad \log_gdp = \log(1 + gdp_country_num).$$

Figure 4 illustrates why this matters by contrasting raw-scale and log-scale variability.

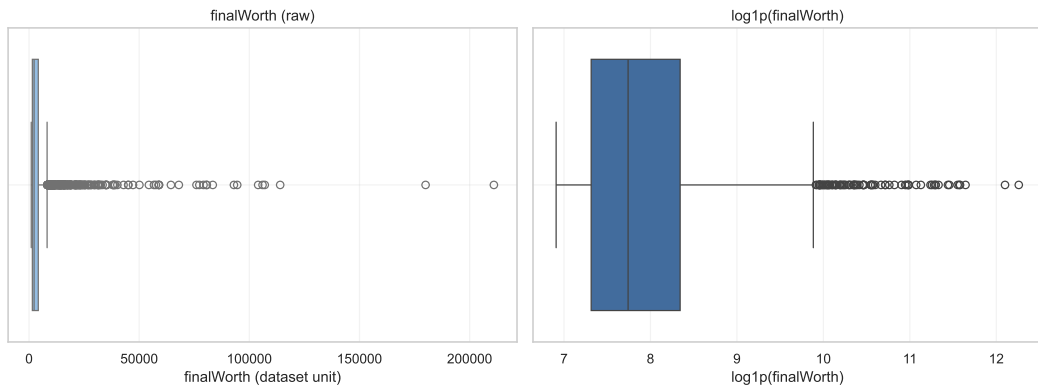


Figure 4: Outliers on raw scale vs. log scale (heavy-tailed behavior).

2.2 Descriptive Statistics

A distribution you should not summarize with “the average”. Figure 5 shows the distribution of billionaire wealth on both raw and log scales. On the raw scale, the plot is dominated by the upper tail; on the log scale, structure becomes visible. Table 2 reports summary statistics that we use throughout the report.

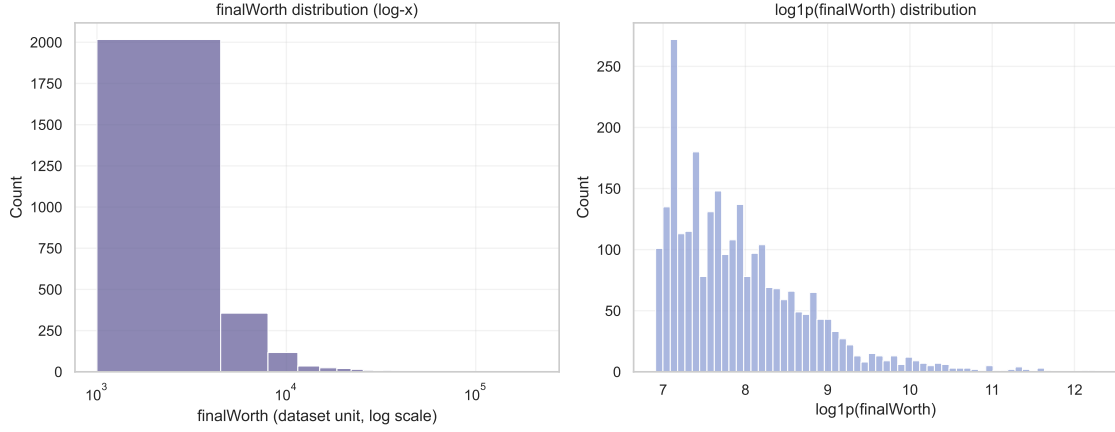


Figure 5: Distribution of `finalWorth` on raw and log scales.

Table 2: Descriptive statistics for `finalWorth`.

stat	finalWorth	log1p(finalWorth)
1%	1000.000000	6.908755
10%	1200.000000	7.090910
25%	1500.000000	7.313887
5%	1100.000000	7.003974
50%	2300.000000	7.741099
75%	4200.000000	8.343078
90%	8000.000000	8.987322
95%	13320.000000	9.497076
99%	41808.000000	10.640328
count	2640.000000	2640.000000
max	211000.000000	12.259618
mean	4623.787879	7.930881
min	1000.000000	6.908755
std	9834.240939	0.820638

Concentration and tail behavior. To visualize tail behavior we use the complementary CDF (CCDF). On a log-log scale, the CCDF highlights the heavy tail without forcing a single parametric distributional assumption. Figure 6 provides this view. As a simple concentration summary, we also compute the wealth share held by the top 1% of records (Figure 7, Table 3).

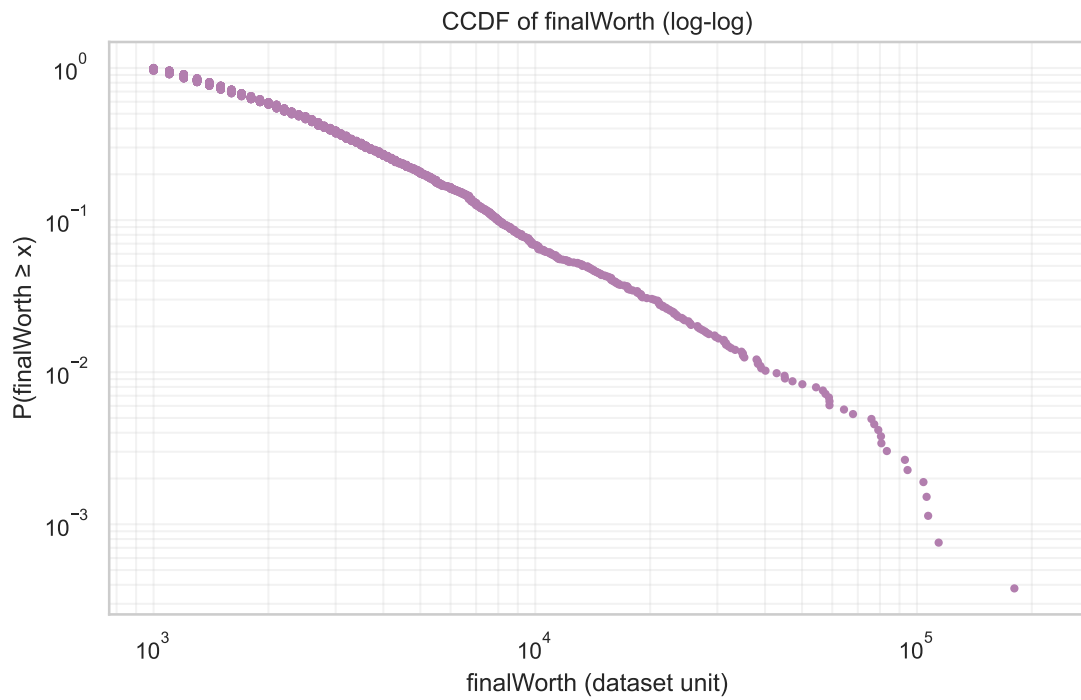


Figure 6: CCDF on log-log scale (heavy-tail visualization).

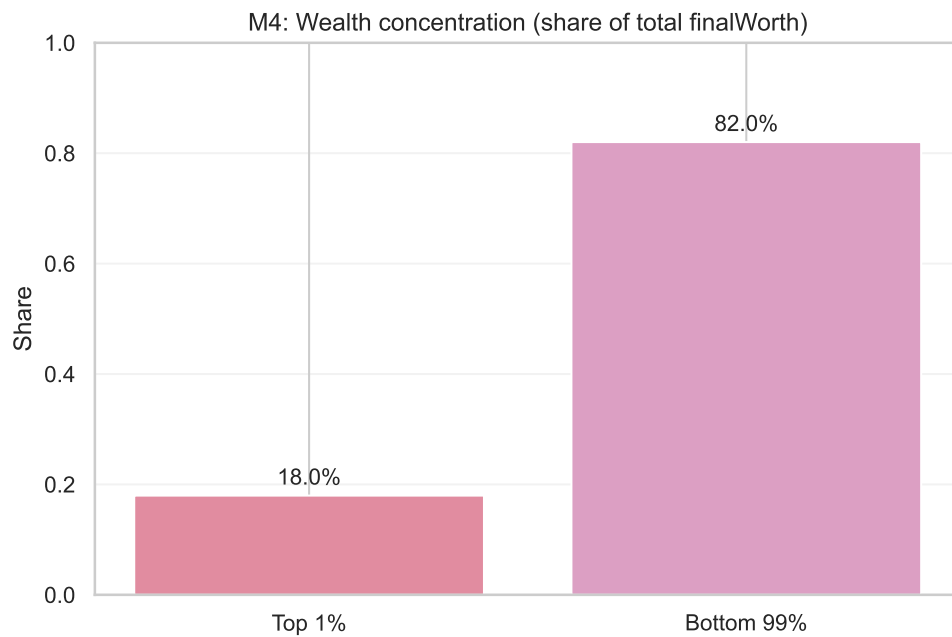


Figure 7: Wealth share of the top 1% of records.

Table 3: Top-1% wealth share (numeric summary).

n_total	top_1pct_n	top_1pct_wealth_share	bottom_99pct_wealth_share
2640	27	0.179801	0.820199

Where and in what industries? Finally, we summarize composition by country and category. These charts are descriptive—they reflect what is recorded in the dataset, not the true global population of billionaires. Figures 8–9 show the top-10 counts.

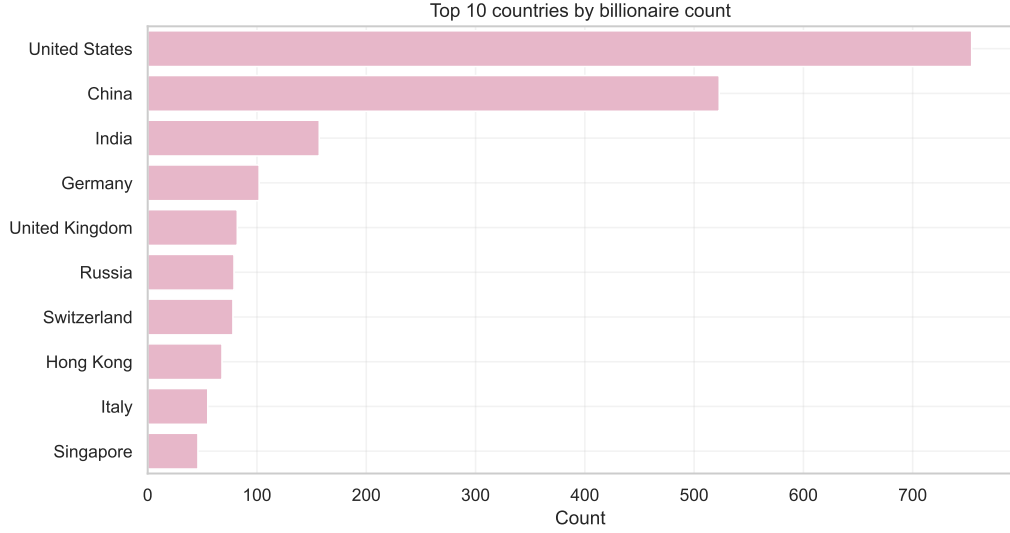


Figure 8: Top-10 countries by billionaire count.

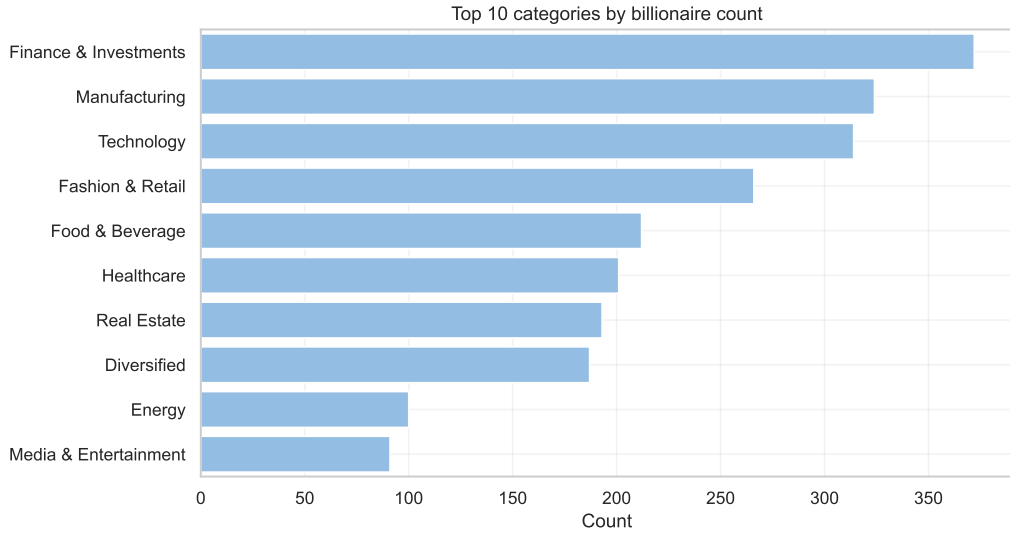


Figure 9: Top-10 categories by billionaire count.

2.3 Inferential Statistics

Why effect sizes (with CIs) are the main object. With thousands of observations, very small differences can be statistically significant. To avoid “p-value storytelling”, we report effect sizes with 95% confidence intervals and include a practical insignificance band in Figure 10. For two-group comparisons, we use: mean difference $\Delta = \bar{x}_1 - \bar{x}_0$, Cohen’s d (standardized mean difference), and Cliff’s δ (stochastic dominance).

For completeness, key definitions are:

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}, \quad d = \frac{\bar{x}_1 - \bar{x}_0}{s_p}, \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_0 - 1)s_0^2}{n_1 + n_0 - 2}},$$

and

$$\delta = P(X_1 > X_0) - P(X_1 < X_0).$$

We compute 95% bootstrap percentile confidence intervals

$$CI_{0.95} = [Q_{0.025}(\hat{\theta}^*), Q_{0.975}(\hat{\theta}^*)],$$

where $\hat{\theta}^*$ denotes bootstrap replicates.

Table 4 reports correlation estimates and group comparisons, and Figure 10 summarizes selected headline effects.

Table 4: Inferential results with effect sizes and 95% bootstrap CIs.

family	analysis	n	effect	ci_low	ci_high	p_value
Correlation	Pearson r (logWorth, logGDP)	2476	0.045759	0.007066	0.082935	0.022786
Correlation	Spearman ρ (logWorth, logGDP)	2476	0.096573	0.057546	0.136319	0.000001
Two-group	Welch t-test: selfMade vs not (logWorth)	2640	-0.097502	-0.168643	-0.031732	0.004980
Two-group	Cohen's d : selfMade vs not (logWorth)	2640	-0.118971	-0.201232	-0.034527	0.004980
Two-group	Cliff's δ : selfMade vs not (logWorth)	2640	-0.079194	-0.124112	-0.031179	0.001072
Robustness	Permutation p-value (mean diff, logWorth)	2640	NaN	NaN	NaN	0.004998
Multi-group	Kruskal-Wallis: logWorth across category	2640	0.008831	NaN	NaN	0.001030

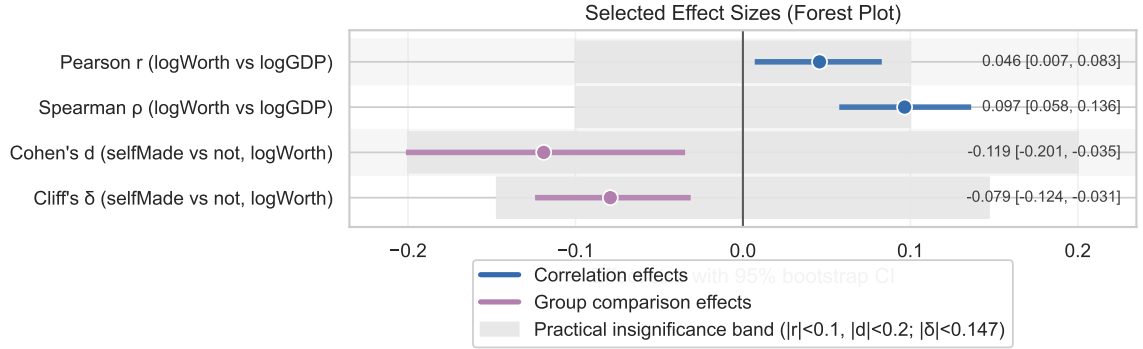


Figure 10: Selected effect sizes with 95% bootstrap confidence intervals. Gray bands indicate “practically small” effects (e.g., $|r| < 0.1$, $|d| < 0.2$).

2.4 Regression Analysis

Model specification (association only). We fit OLS models to quantify the association between wealth and GDP. On the log scale:

$$\log_finalWorth_i = \beta_0 + \beta_1 \log_gdp_i + \varepsilon_i.$$

Interpreting β_1 as “elasticity-like” is convenient (a change in log-wealth per unit change in log-GDP), but it is not a causal effect.

Robust inference and model checking. Given heavy tails and likely heteroskedasticity, we report HC3 robust standard errors and 95% confidence intervals. Regression coefficients are summarized in Table 5. Figure 11 visualizes the fitted association, and Figure 12 provides basic diagnostics. We include the Breusch–Pagan test as a heteroskedasticity diagnostic (Table 6).

Table 5: Regression results (OLS with HC3 robust SE and 95% CI).

model	term	coef	se_hc3	
A: finalWorth ~ gdp_country_num (HC3)	const	4237.583714	281.536968	15.05
A: finalWorth ~ gdp_country_num (HC3)	gdp_country_num	0.000000	0.000000	1.87
B: log_finalWorth ~ log_gdp (HC3)	const	7.266857	0.283538	25.62
B: log_finalWorth ~ log_gdp (HC3)	log_gdp	0.022871	0.009707	2.35
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	Intercept	6.387722	0.350104	18.24
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	C(category)[T.Fashion & Retail]	0.157747	0.086165	1.83
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	C(category)[T.Finance & Investments]	0.118336	0.077501	1.52
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	C(category)[T.Food & Beverage]	0.076832	0.089285	0.86
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	C(category)[T.Healthcare]	-0.096619	0.081598	-1.18
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	C(category)[T.Manufacturing]	-0.097177	0.077834	-1.24
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	C(category)[T.Real Estate]	-0.085950	0.080119	-1.07
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	C(category)[T.Technology]	0.207634	0.086833	2.39
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	log_gdp	0.035423	0.011781	3.00
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	age	0.009192	0.001517	6.05
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	selfMade_bin	-0.192678	0.044844	-4.29
C: log_finalWorth ~ log_gdp + controls + C(category) (HC3)	gender_F	-0.040499	0.059107	-0.68

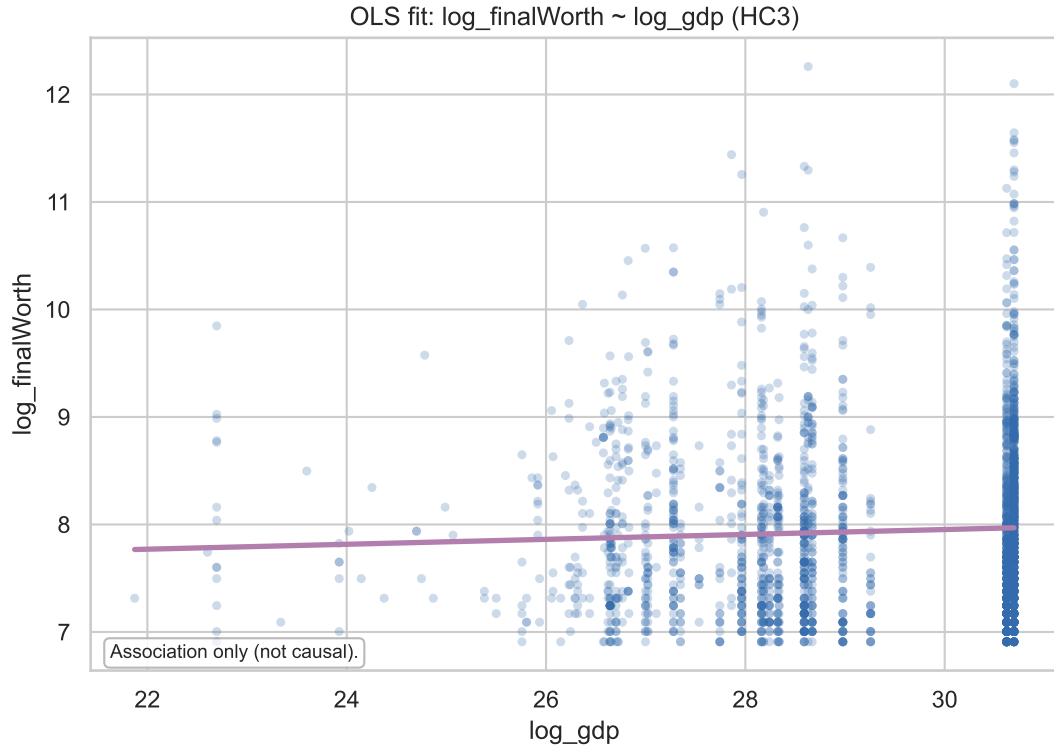


Figure 11: OLS fit for log_finalWorth ~ log_gdp (association only).

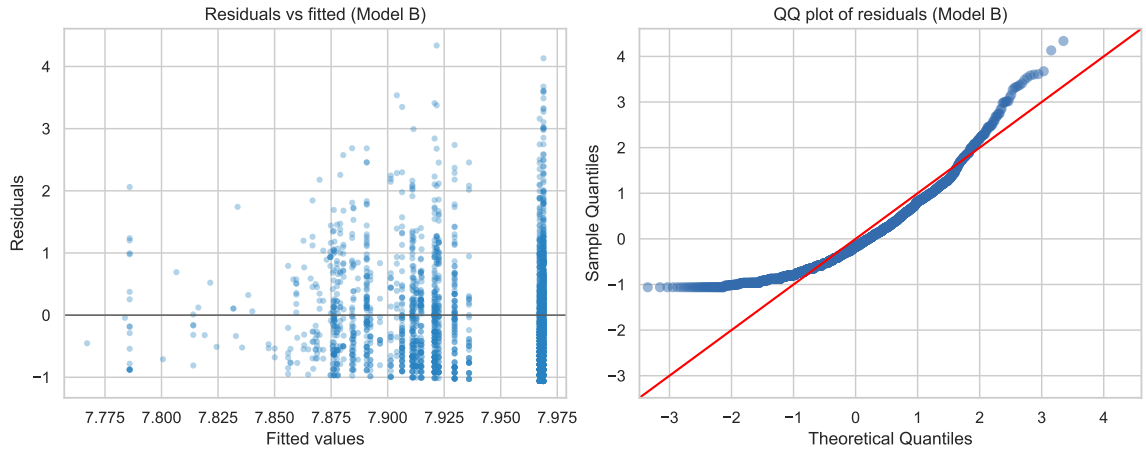


Figure 12: Diagnostics for the main log-log model.

Table 6: Breusch–Pagan test for heteroskedasticity (Model B residuals).

lm_stat	lm_pvalue	f_stat	f_pvalue	note
2.221610	0.136091	2.221809	0.136201	Breusch–Pagan test for heteroskedasticity (Model B residuals)

Robustness: does the result survive when we remove the extreme tail? A fair critique is that the estimated association could be driven by a tiny number of the largest fortunes. We therefore re-fit the key log model after dropping the top 1% by `finalWorth`. Table 7 and Figure 13 show the stability of β_1 under this stress test.

Table 7: Robustness: coefficient on `log_gdp` with and without top-1% exclusion.

setting	n	coef_log_gdp	ci_low	ci_high
All complete cases	2476	0.022871	0.003836	0.041906
Drop top 1% <code>finalWorth</code>	2451	0.013569	-0.004519	0.031658

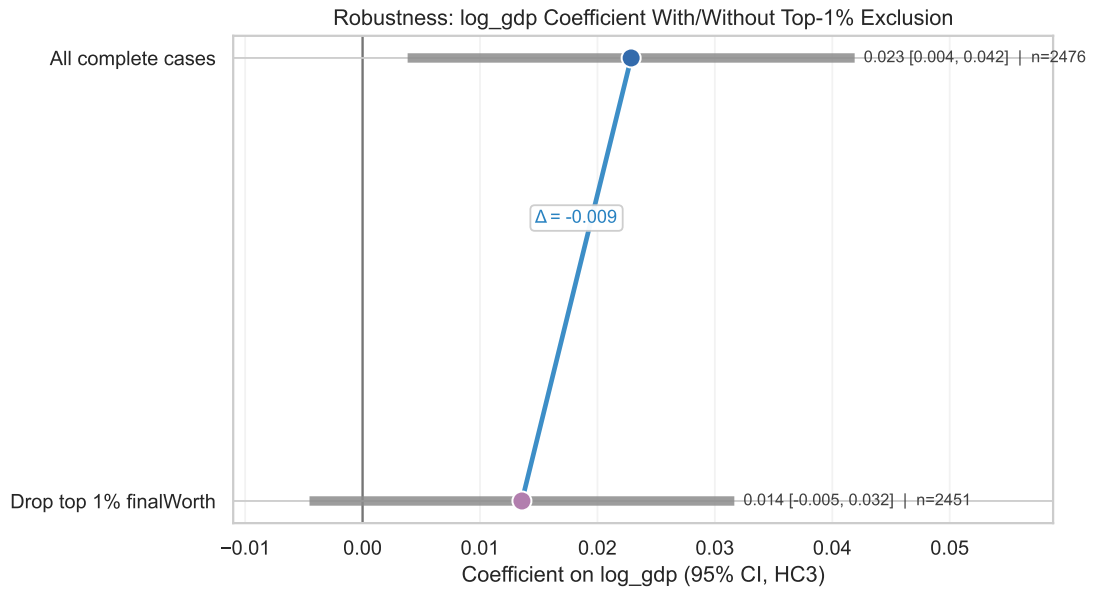


Figure 13: Robustness: connected dumbbell comparison of β_1 across settings (95% CI).

3 Key Insights and Limitations

This section presents the story in its final form: four takeaways (two core, two exploratory), each tied to concrete evidence and at least one robustness consideration.

3.1 Core Insight 1: Billionaire wealth is dominated by the upper tail

The dataset behaves like many “extreme outcome” datasets: the main challenge is not estimating the center, but understanding the tail. On the raw scale (Figure 5), a handful of observations visually dominate the distribution; on the log scale, the rest of the data becomes visible. The CCDF (Figure 6) makes the same point from a different angle: the tail decays slowly, so small changes in how we handle extremes can change results. **Design response:** we report key results on a log scale and avoid interpreting raw-scale averages as “typical” wealth.

3.2 Core Insight 2: Wealth–GDP association is present but should be read as non-causal

Across countries in this dataset, higher GDP is associated with higher recorded billionaire wealth. The regression table (Table 5) and fitted plot (Figure 11) summarize this relationship on the log scale. However, this is an observational association: GDP may proxy institutions, market size, reporting intensity, and historical accumulation. **Design response:** inference uses HC3 robust SE, diagnostics are reported (Figure 12, Table 6), and we stress-test the key coefficient by excluding the top 1% of fortunes (Table 7, Figure 13).

3.3 Exploratory Insight 1: Missingness is structured, not random

A story that ignores missingness can easily become fiction. Here, the state/region fields are essentially U.S.-only (Figure 1), which means those variables cannot be treated as universally observed. Country-dependent missingness patterns also appear in the heatmap (Figure 3). **Design response:** we either restrict those variables to the relevant subset (U.S. records) or keep them out of global models, and we interpret cross-country patterns as “recorded in this dataset” rather than population truth.

3.4 Exploratory Insight 2: Effect sizes help separate “real” from “important”

Because the dataset is large, some effects can be statistically detectable even when practically small. Figure 10 adds a practical insignificance band to keep interpretation grounded (e.g., $|r| < 0.1$ and $|d| < 0.2$). Table 4 reports the full set of effect sizes with bootstrap CIs, which is more informative than a p -value-only summary. **Design response:** we discuss effect size magnitudes and uncertainty, not just significance.

3.5 Limitations

- **Observational data:** no causal claims are made. GDP is correlated with many unmeasured confounders.
- **Dataset coverage:** the dataset is curated and may reflect reporting/collection practices; composition charts are descriptive of the dataset, not the world.
- **Documentation gaps:** when variable units or definitions are not fully documented, we avoid over-precise interpretations and focus on internally consistent comparisons.

4 References

- Course-provided dataset: *Billionaires Statistics Dataset.csv*.
- Python scientific stack: pandas, numpy, scipy, statsmodels, matplotlib (versions recorded in Table 8).

A Reproducibility and Notebook Appendix

Table 8: Environment snapshot (for reproducibility).

timestamp	python	platform	numpy	pandas	matplotlib	seaborn
2026-02-28T02:13:50	3.13.5	Windows-11-10.0.26100-SP0	2.1.3	2.2.3	3.10.0	0.13.2

Table 9: Runbook for reproducing all artifacts.

step	action
1	Place the raw CSV at ./data/raw/Billionaires Statistics Dataset.csv
2	Open ./code/project_stastistic_analyse_and_visualization.ipynb
3	Kernel → Restart & Run All
4	Verify outputs in ./output/fig and ./output/tab
5	Use tab_M9_report_map to reference figures/tables in the report

Table 10: Report map: where each figure/table is used.

report_section	module	kind	name	paths
Introduction	M0	figure	fig_M0_variable_role_counts	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Introduction	M0	table	tab_M0_environment_snapshot	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Introduction	M0	table	tab_M0_research_questions	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Introduction	M0	table	tab_M0_research_questions_source	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Introduction	M0	table	tab_M0_schema_freeze_A_summary	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Introduction	M0	table	tab_M0_variable_roles	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Introduction (Dataset overview)	M1	figure	fig_M1_dtype_distribution	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Introduction (Dataset overview)	M1	table	tab_M1_dictionary_verification_checks	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Introduction (Dataset overview)	M1	table	tab_M1_gdp_parse_summary	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Introduction (Dataset overview)	M1	table	tab_M1_schema_summary	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Data Quality)	M2	figure	fig_M2_missingness_heatmap_country_top12	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Data Quality)	M2	figure	fig_M2_missingness_top15	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Data Quality)	M2	figure	fig_M2_outliers_raw_vs_log_box	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Data Quality)	M2	figure	fig_M2_us_only_state_coverage	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Data Quality)	M2	table	tab_M2_artifact_index	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Data Quality)	M2	table	tab_M2_dq_issues_summary	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Data Quality)	M2	table	tab_M2_missingness_by_country_top12	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Data Quality)	M2	table	tab_M2_missingness_top15	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Data Quality)	M2	table	tab_M2_us_only_state_coverage	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Cleaning)	M3	figure	fig_M3_validation_log_distributions	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Cleaning)	M3	table	tab_M3_clean_dataset_manifest	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Cleaning)	M3	table	tab_M3_cleaning_log	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Descriptive)	M4	figure	fig_M4_finalWorth_ccdf_loglog	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Descriptive)	M4	figure	fig_M4_finalWorth_distribution_raw_and_log	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Descriptive)	M4	figure	fig_M4_top10_category_count	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Descriptive)	M4	figure	fig_M4_top10_country_count	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Descriptive)	M4	figure	fig_M4_top1pct_wealth_share	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Descriptive)	M4	table	tab_M4_finalWorth_describe	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Descriptive)	M4	table	tab_M4_top10_category_by_count	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Descriptive)	M4	table	tab_M4_top10_country_by_count	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Descriptive)	M4	table	tab_M4_top1pct_wealth_share	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
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Statistical Analysis (EDA)	M5	figure	fig_M5_box_logWorth_by_selfMade	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (EDA)	M5	figure	fig_M5_corr_heatmap_spearman	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (EDA)	M5	figure	fig_M5_scatter_finalWorth_vs_gdp_raw	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (EDA)	M5	figure	fig_M5_scatter_logWorth_vs_logGDP	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (EDA)	M5	figure	fig_M5_scatter_logWorth_vs_logGDP_by_selfMade	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (EDA)	M5	table	tab_M5_corr_matrix_spearman	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (EDA)	M5	table	tab_M5_groupwise_correlations	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Inferential)	M6	figure	fig_M6_effectsize_forest	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Inferential)	M6	table	tab_M6_inferential_results	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Inferential)	M6	table	tab_M6_posthoc_category_pairwise	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Regression)	M7	figure	fig_M7_regression_diagnostics	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Regression)	M7	figure	fig_M7_regression_fit_loglog	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Regression)	M7	figure	fig_M7_robust_coef_compare_top1pct	{"pdf": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Regression)	M7	table	tab_M7_bp_test_modelB	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Regression)	M7	table	tab_M7_regression_main	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Statistical Analysis (Regression)	M7	table	tab_M7_robust_drop_top1pct_logmodel	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\
Key Insights & Limitations (Robustness)	M8	table	tab_M8_robustness_matrix	{"csv": "E:\\canfiles\\Common courses study\\u5927u5b66\\

Notebook appended. The full analysis notebook is included as ./code/project_stastistic_analyse_and_visualization. All figures and tables referenced above are generated by running the notebook from top to bottom.